THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 1: **Chemical Reactions and Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The balanced equation for burning of magnesium ribbon in air is: 1

(a) $2\,Mg + O_2 \longrightarrow 2\,MgO_2$ **(b)** $Mg + O_2 \longrightarrow MgO$ **(c)** $2\,Mg + O_2 \longrightarrow 2\,MgO$ **(d)** $Mg + O \longrightarrow MgO$

2. Which of the following is a double displacement reaction? 1

(a) $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$ **(b)** $CaCO_3 \longrightarrow CaO + CO_2$ **(c)** $C + O_2 \longrightarrow CO_2$ **(d)** $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$

3. $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ is a displacement reaction because: 1

(a) zinc is less reactive than copper
(b) copper gets oxidised
(c) both metals are exchanged as anions
(d) zinc is more reactive and displaces copper from its salt

4. In the reaction $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$, the type of reaction is: 1

(a) Displacement **(b)** Combination
(c) Double displacement **(d)** Decomposition

5. The white precipitate formed on mixing sodium sulphate and barium chloride solutions is: 1

(a) $BaSO_4$ **(b)** $BaCl_2$ **(c)** Na_2SO_4 **(d)** $NaCl$

6. After balancing $H_2 + Cl_2 \longrightarrow HCl$, the coefficient of HCl is: 1

(a) 1 **(b)** 3 **(c)** 4 **(d)** 2

7. A solution of substance X is used for whitewashing. X is: 1

(a) Calcium oxide (quick lime) **(b)** Calcium carbonate
(c) Calcium sulphate **(d)** Sodium hydroxide

8. Oxidation is the: 1

(a) loss of oxygen or gain of hydrogen **(b)** gain of oxygen or loss of hydrogen
(c) gain of both oxygen and hydrogen **(d)** loss of both oxygen and hydrogen

9. In the electrolysis of water, the volumes of hydrogen and oxygen collected are in the ratio: 1

(a) 1 : 2 **(b)** 3 : 1 **(c)** 2 : 1 **(d)** 1 : 1

10. Magnesium ribbon is cleaned before burning in air in order to: 1

(a) make it burn with a green flame
(b) make it heavier
(c) remove the protective oxide layer from its surface
(d) convert it into a powder

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why should a magnesium ribbon be cleaned before it is burnt in air? 2

12. Why do we apply paint on iron articles? 2

13. What is observed when zinc granules are added to copper sulphate solution? Name the type of reaction. 2

14. Write the balanced chemical equation for the reaction of hydrogen with chlorine to give hydrogen chloride. 2

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SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why does the colour of copper sulphate solution change when an iron nail is dipped in it? Write the equation. **3**

16. What is a balanced chemical equation? Balance $\mathrm{Fe + H_2O \longrightarrow Fe_3O_4 + H_2}$. **3**

17. A solution of substance X is used for whitewashing. Name X and write its reaction with water. **3**

OR

Give an example of a double displacement reaction other than the one between sodium sulphate and barium chloride. Write its equation and the observation. **3**

18. Differentiate between a combination and a decomposition reaction with one example each. **3**

OR

In the electrolysis of water, why is the volume of gas collected in one test tube double that in the other? Name the gases. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. What is a balanced chemical equation? Balance $\mathrm{Fe + H_2O \longrightarrow Fe_3O_4 + H_2}$ step by step and write state symbols. **5**

OR

Define oxidation and reduction. Identify the substance oxidised, reduced, the oxidising agent and the reducing agent in $\mathrm{ZnO + C \longrightarrow Zn + CO}$. **5**

20. Describe what happens when a magnesium ribbon is burnt in air. Write the equation, name the type of reaction and state the nature of the product. **5**

OR

Explain how corrosion and rancidity affect our daily life. Give two prevention methods for each. **5**

END OF QUESTION PAPER — **Total: 40 Marks**

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